PRACTICAL LAB WRITE-UP: SOC ALERT INVESTIGATION

Week 5 — Checking Alerts (The Tier 1 SOC Job) | DG Interns Hub

PRACTICAL LABORATORY METADATA			
Author / Analyst:	SOC Analyst Intern	**Organization:**	DG Interns Hub
SIEM Platform:	Splunk Enterprise 10.6.1	**Target OS:**	Ubuntu Linux 6.x (Oracle VirtualBox)
Monitored Host:	`ubuntu-soc`	**Log Source:**	`/var/log/auth.log`
Splunk Index:	`main`	**Sourcetype:**	`linux_secure`
Execution Date:	November 6, 2026	**Status:**	Submitted / Complete

DOCUMENT STRUCTURE & TABLE OF CONTENTS

Section 1: Lab Overview & Objectives
Section 2: Lab Infrastructure & Environment
Section 3: Splunk Log Ingestion & Validation
Section 4: Seven Detection Searches (01 - 07)
Section 5: Alert Investigation & Triage
Section 6: Threat Intelligence Enrichment
Section 7: MITRE ATT&CK Matrix Mapping
Section 8: Key Learnings & Conclusion

1. LAB OVERVIEW

Title: Week 5 – Checking Alerts (The L1 SOC Job)

Objective: The objective of this practical laboratory write-up is to document the process by which a Level 1 (L1) Security Operations Center (SOC) Analyst triages security alerts, validates suspicious system events, identifies benign operational false positives, reviews authentication patterns, and documents findings using an enterprise SIEM platform (Splunk Enterprise 10.6.1).

This hands-on assignment focuses on executing seven (7) targeted Search Processing Language (SPL) queries against ingested Ubuntu Linux authentication telemetry, mapping findings to MITRE ATT&CK tactics, and maintaining analytical rigor by separating verified log evidence from telemetry gaps.

2. LAB ENVIRONMENT

The practical lab was constructed within a virtualized security sandbox using Oracle VirtualBox running Ubuntu Linux as the endpoint target and Splunk Enterprise 10.6.1 as the centralized SIEM.

COMPONENT	LAB SPECIFICATION	OPERATIONAL ROLE IN SOC WORKFLOW
SIEM Engine	Splunk Enterprise `10.6.1`	Centralized log collection, indexing, SPL parsing, search correlation, and alert display interface.
Operating System	Ubuntu Linux (Kernel 6.x)	Target guest endpoint operating system generating host authentication and process logs.
Hypervisor	Oracle VirtualBox	Isolated virtualization environment hosting the `ubuntu-soc` virtual machine.
Log Source Path	`/var/log/auth.log`	System log file recording PAM events, SSH logins, `sudo` commands, and user account creation.
Index & Sourcetype	`index=main` / `linux_secure`	Splunk repository index and sourcetype definition parsing Linux authentication syslog formatting.

3. SPLUNK LOG INGESTION & VALIDATION

To establish SOC visibility, local Linux authentication logs (`/var/log/auth.log`) were onboarded into Splunk Enterprise via file monitoring input settings:

- **Ingestion Configuration:** Splunk was configured to monitor `/var/log/auth.log`, assigning indexed data to `index=main` and explicitly setting `sourcetype=linux_secure`.
- **Validation Process:** Ingestion was verified by executing `index=main sourcetype=linux_secure | head 10` to confirm timestamps, hostname parsing (`ubuntu-soc`), and raw message field extraction.
- **SOC Value of Authentication Logs:** Linux authentication logs provide critical visibility into access vector hygiene, credential misuse, local privilege escalation, interactive shell sessions, and unauthorized user account persistence.

4. SEVEN DETECTION SEARCHES (DETAILED ANALYSIS)

SEARCH 01 — Brute Force Detection

Objective: Detect repeated local or remote authentication failures clustered by host, process, and target account.

MITRE ATT&CK Mapping: `T1110 — Brute Force` (Sub-technique: `T1110.001 — Password Guessing`)

```
index=main sourcetype=linux_secure ("authentication failure" OR "Failed password") | stats
count as failed_attempts by host, process, logname | sort - failed_attempts
```

SPL Command Breakdown: `index=main sourcetype=linux_secure` filters Linux auth logs; string matching isolates failure events; `stats count by...` aggregates attempts across host, PAM process, and user; `sort -` presents highest failure volumes first.

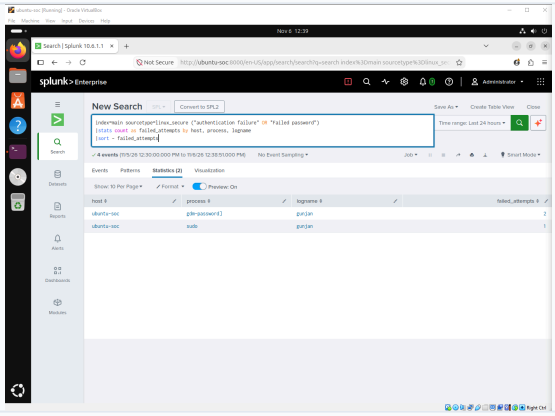


Figure 1: Splunk Search 01 Evidence — 4 raw events evaluated into 2 aggregated failure rows for user gunjan.

- **Actual Findings:** 4 raw events evaluated into 2 statistical summary rows. Target: user `gunjan` on host `ubuntu-soc`. Processes: `gdm-password` (2 failures) and `sudo` (1 failure).
- **L1 Analyst Interpretation:** Low volume failure cluster. Failures occurred locally via desktop GUI unlock (`gdm-password`) and local privilege escalation (`sudo`). No remote SSH (`sshd`) or external source IP was recorded. Activity is consistent with human typing error during local operation.
- **Limitations & Next Steps:** Cannot detect network-level port scanning or remote SSH brute-forcing if SSHD is unmonitored. Next steps: Check for successful login immediately following failure window.

SEARCH 02 — New Account Creation

Objective: Identify the creation of new local user accounts on Linux endpoints.

MITRE ATT&CK Mapping: `T1136 — Create Account` (Sub-technique: `T1136.001 — Local Account`)

```
index=main sourcetype=linux_secure ("useradd" OR "adduser") | table _time host _raw | sort -
_time
```

SPL Command Breakdown: Filters for Linux binary execution strings `useradd` or `adduser`; `table` formats output fields; `sort - _time` orders events chronologically.

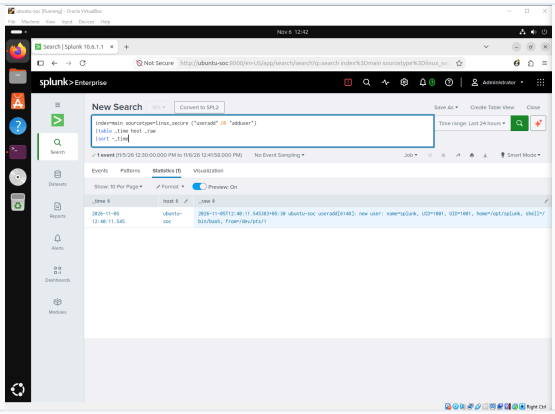


Figure 2: Splunk Search 02 Evidence — 1 event detecting useradd creation of user account 'splunk'.

- **Actual Findings:** 1 event returned at `2026-11-05 12:40:11.545 IST`. Details: New user `splunk`, UID 1001, GID 1001, Home `/opt/splunk`, Shell `/bin/bash` from pseudoterminal `/dev/pts/1`.
- **L1 Analyst Interpretation:** Account creation requires scrutiny for persistence. However, creating account `splunk` with home directory `/opt/splunk` directly aligns with administrative setup of Splunk Enterprise during initial lab installation.
- **Limitations & Next Steps:** Does not reveal permissions granted post-creation. Next steps: Cross-reference Change Management tickets for authorized installer activity.

SEARCH 03 — Suspicious PowerShell Execution

Objective: Detect Windows PowerShell process creation and command-line execution.

MITRE ATT&CK Mapping: `T1059.001 — Command and Scripting Interpreter: PowerShell`

```
index=* (EventCode=1 OR EventID=1) | search Image="*powershell.exe*" | table _time host User Image CommandLine | sort - _time
```

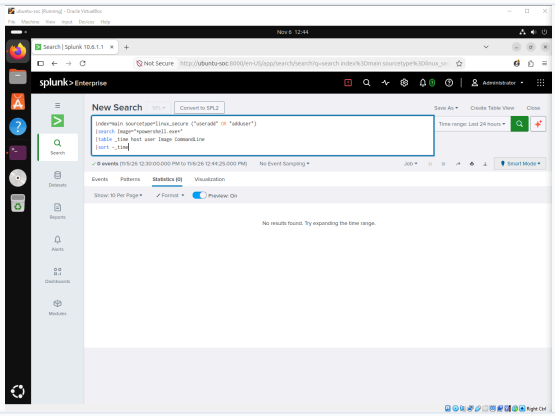


Figure 3: Splunk Search 03 Evidence — 0 events returned due to lack of Windows Sysmon process telemetry.

- **Actual Findings:** 0 events returned ("No results found").
- **L1 Telemetry Gap Analysis:** The lab dataset exclusively ingests Ubuntu Linux authentication logs (`sourcetype=linux_secure`). It lacks Windows Sysmon process creation telemetry (Event ID 1). **Zero results do not prove PowerShell activity is absent** from the enterprise; it reflects a telemetry gap.

SEARCH 04 — Network Service Scanning

Objective: Identify network port scanning, SYN sweeps, or Nmap reconnaissance tools.

MITRE ATT&CK Mapping: `T1046 — Network Service Discovery`

```
index=* ("SYN Scan" OR "port scan" OR "Nmap") | table _time host source src_ip dest_ip _raw |
sort - _time
```

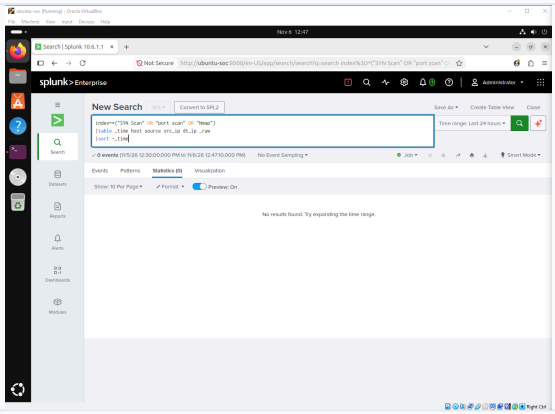


Figure 4: Splunk Search 04 Evidence — 0 events returned due to lack of Network NIDS / Firewall telemetry.

- **Actual Findings:** 0 events returned ("No results found").
- **L1 Telemetry Gap Analysis:** Network IDS/IPS (Suricata/Zeek) and firewall logs are not onboarded. Reconnaissance port scans cannot be confirmed or ruled out using host authentication logs alone.

SEARCH 05 — Phishing-related Activity

Objective: Detect phishing indicators, suspicious mail attachments, or mail gateway alerts.

MITRE ATT&CK Mapping: `T1566 — Phishing`

```
index=* ("phishing" OR "phish" OR "suspicious attachment") | table _time host source _raw |
sort - _time
```

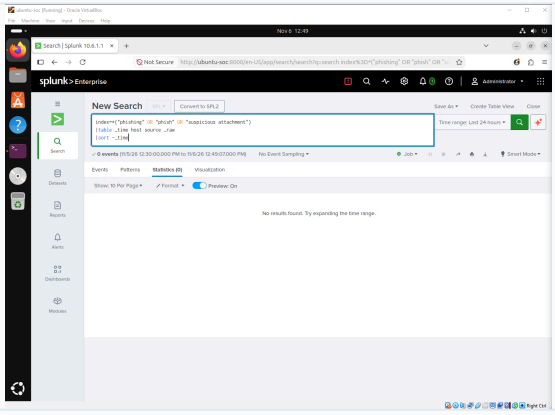


Figure 5: Splunk Search 05 Evidence — 0 events returned due to lack of Email Gateway telemetry.

- **Actual Findings:** 0 events returned ("No results found").
- **L1 Telemetry Gap Analysis:** Dataset lacks Secure Email Gateway (SEG) or Exchange/O365 logs. Phishing threat vectors cannot be evaluated via Linux authentication telemetry.

SEARCH 06 — Linux Authentication Failure Analysis

Objective: Group authentication failures by process values to analyze multi-vector login failures.

MITRE ATT&CK Mapping: `T1110 — Brute Force` (Potential Analytical Mapping)

```
index=main sourcetype=linux_secure ("authentication failure" OR "Failed password") | stats
count as failure_count values(process) as processes by host, logname | sort - failure_count
```

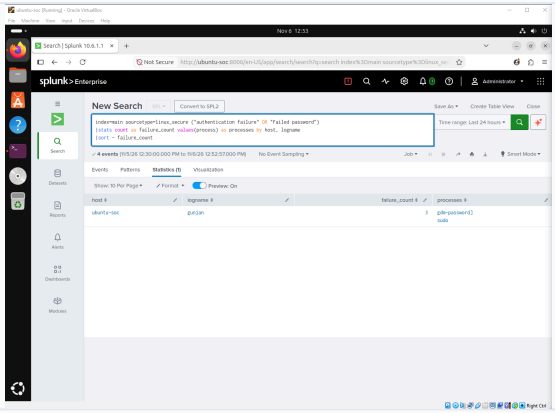


Figure 6: Splunk Search 06 Evidence — 4 raw events grouped into 1 aggregated summary row (3 failures).

- **Actual Findings:** 4 raw events evaluated into 1 consolidated row. Host: `ubuntu-soc`, User: `gunjan`, Failure Count: 3, Processes: `gdm-password` and `sudo`.
- **Analytical Distinction & Overlap with Search 01:** Searches 01 and 06 evaluate the *exact same underlying raw log data*. Search 06 uses `values(process)` to group processes into a multi-value field. They represent **two analytical presentations of the same local event**, not two separate attacks.

SEARCH 07 — Linux Privileged Command Execution

Objective: Monitor execution of administrative commands using `sudo` or `pkexec`.

MITRE ATT&CK Mapping: N/A (Generic Administrative Monitoring)

```
index=main sourcetype=linux_secure "COMMAND=" | table _time host _raw | sort - _time
```

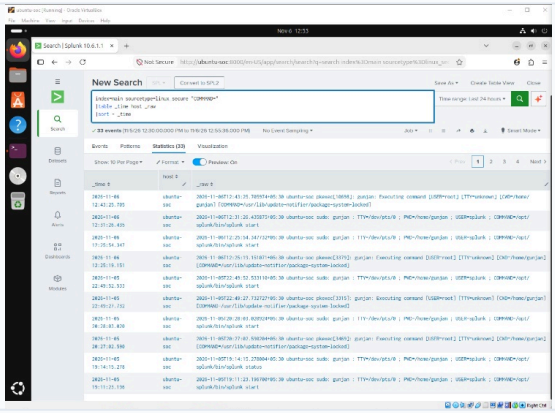


Figure 7: Splunk Search 07 Evidence — 33 privileged execution events across 4 result pages.

- **Actual Findings:** 33 total events across 4 pagination pages. User: `gunjan`. Commands include `/opt/splunk/bin/splunk start`, `status`, and `pkexec package-system-locked`.
- **L1 Analyst Interpretation:** Privileged execution monitoring is essential. Review confirms commands reflect routine administrative management of Splunk Enterprise and normal desktop updates. Activity is non-malicious.